

Sex effect on Vasopressor use after non-cardiac Surgery

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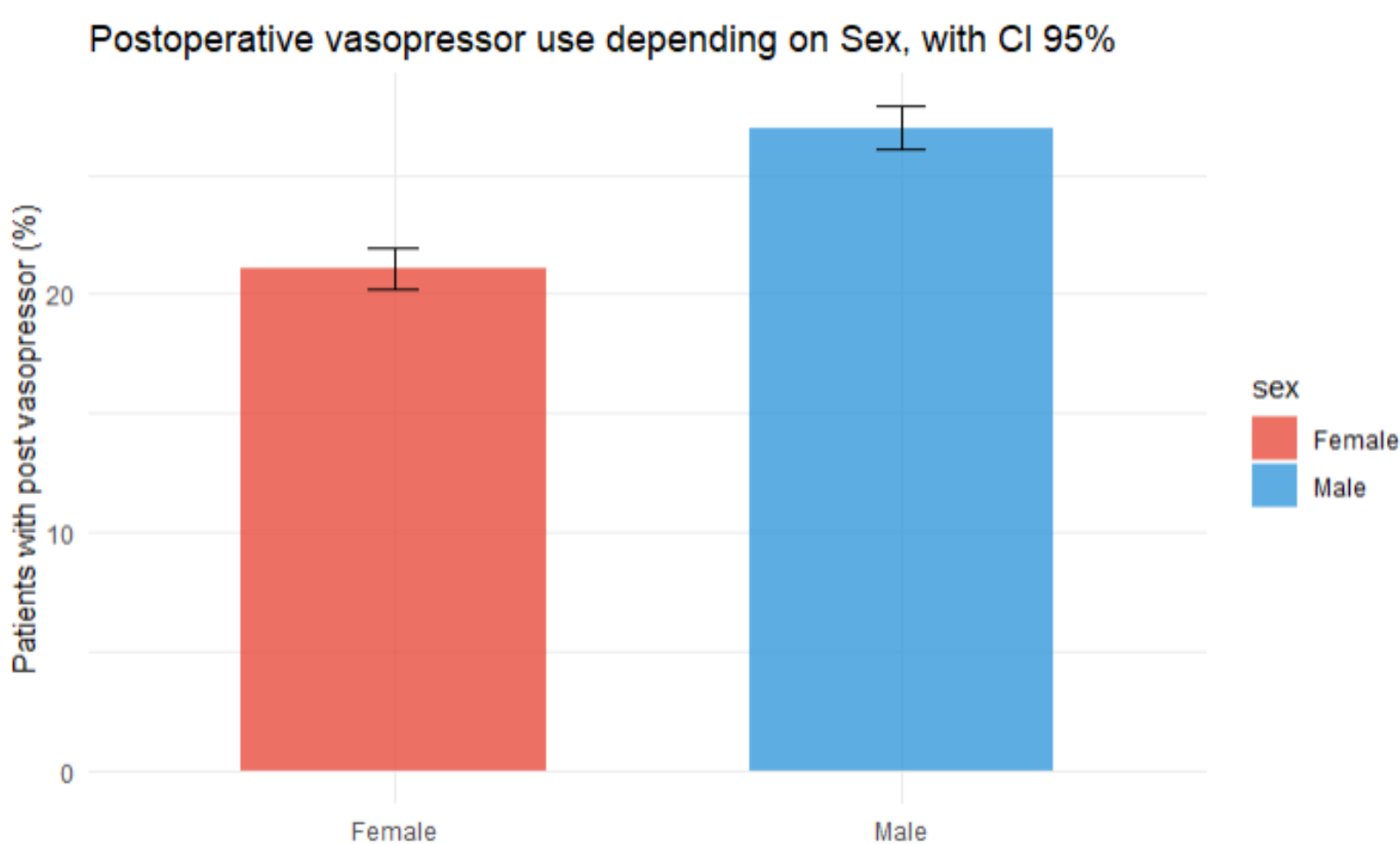
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1. Introduction

- A study concerning non-cardiac surgery revealed that 3.9% of patients receive post-vasopressor treatment.
- This group experiences a heightened mortality rate.
- Male patients exhibit a higher likelihood of requiring postoperative vasopressor support compared to female patients.
- This study aims to systematically investigate the clinical and biological factors contributing to the higher vasopressor requirement observed in male patients after non-cardiac surgery.

2. Definition



- Vasopressors are medications used to elevate arterial pressure by constricting blood vessels (often required after surgery).
- Postoperative vasopressors are administered after a surgery, and only a specific subset of patients requires them.
- In our dataset, approximately 22% of women and 27% of men used postoperative vasopressors. We aim to explain the drivers of this disparity.

3. Methods

- **Study Population:** We analyzed 18,387 non-cardiac surgical cases from the SQUEEZE dataset, excluding cases with missing values.
- **Statistical Modeling:** We employed mixed-effects logistic regression models (adjusting for age, ASA score, urgency, and procedure) with random effects for countries and hospitals.
- **Machine Learning:** A Random Forest algorithm was used to identify variable importance without assuming linearity.
- **Validation:** We utilized a mediation analysis (for fluid volumes), 1:1 Propensity-Score Matching (PSM), and E-value tests to isolate intrinsic biological effects and evaluate unmeasured confounding.

4. Key Findings: Fluids and Procedures

- **The Fluid Standardization Paradox:** Fluid administration is standardized by weight (mean 15 mL/kg for males and 14 mL/kg for females). However, a mediation analysis revealed that weight-based protocols do not explain the sex gap. A direct biological effect of 4.7 percentage points persists for males even when fluid volumes per kilogram are mathematically equalized.
- **Surgical Complexity vs. Biology:** In hepato-biliary and vascular procedures, the male excess is entirely driven by higher case complexity (ASA, blood loss). Conversely, a genuine biological residual effect persists in upper gastro-intestinal and neurological procedures.

5. Results

Characteristic	Overall	Female	Male
N	18,385	8,795	9,590
Median Weight (kg)	75	70	80
Urgent Surgery (%)	64%	65%	62%
Postop Vasopressors (%)	24%	21%	27%
Lowest intraop MAP (mmHg)	64	63	65

Table 1: Baseline characteristics of the study population stratified by sex.

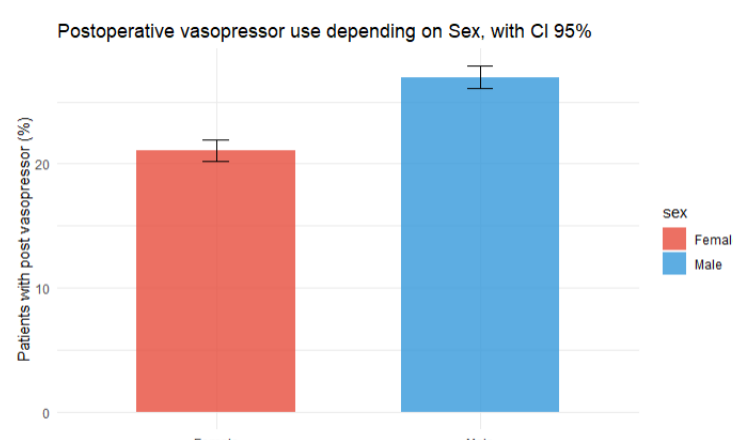


Figure 2: Odds Ratios for vasopressor use by procedure. Male excess varies significantly across surgical contexts.

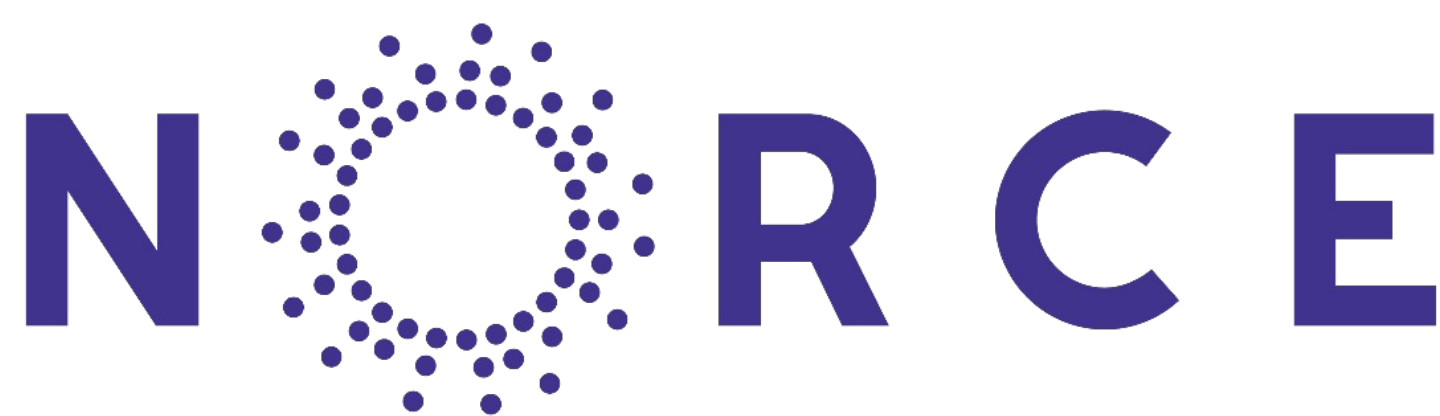
6. The Spinal Exception & Age

- **A Striking Reversal:** While males are generally at higher risk, spinal anesthesia constitutes a notable exception. In a propensity-matched cohort, females had a 4-fold higher risk of receiving vasopressors under spinal anesthesia compared to matched males ($OR = 0.25$, 95% CI [0.15–0.38]).
- **Age Convergence:** The risk gap between genders is maximal in younger patients but narrows significantly after age 70. In the elderly cohort, vascular stiffness and geriatric frailty become the dominant drivers of instability, overriding initial female biological protections.

7. Conclusions

- The male excess in postoperative vasopressor requirement is a statistically robust phenomenon that cannot be attributed to differences in surgical complexity, institutional practice, or simple procedural selection.
- **The "Double Hit":** Male patients suffer from a sharper pre-induction MAP drop (acute stress response) combined with a relative under-filling from strict weight-based fluid protocols (mL/kg).
- **Clinical Implications:**
 - Hemodynamic management should be personalized, moving beyond strict weight-based standards to account for gender-specific physiological trajectories.
 - Vasopressor initiation criteria and fluid preloading thresholds may need to be differentiated by sex, particularly for spinal procedures.

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